

CML NRZ Driver Design

65 nm CMOS Current-Mode Logic Driver for 12.5 Gb/s NRZ Links

Design, AC & Transient Verification, and Eye-Diagram Characterization

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1. Objective

The goal of this work is to design a Current-Mode Logic (CML) NRZ line driver in 65 nm CMOS capable of operating up to 12.5 Gb/s. The transmitter must comply with the specifications defined for a DC-coupled link, summarized in Table 1.

Table 1 — Target Specifications for the CML Driver (DC-Coupled Link) **Table 1 — Target Specifications for the CML Driver (DC-Coupled Link)**

Symbol	Supply / Parameter	Value
Predriver Supply	VDD (predriver)	1 V
Driver Supply	VDD (driver)	1.2 V
Unit Interval (UI)	1 / (12.5 Gb/s)	80 ps
Tx Differential Voltage	Into 100 Ω load	600 mVppd
Rx Differential Load	Floating, between OUTP/OUTN	100 Ω

At 12.5 Gb/s, the unit interval is $UI = 1/(12.5 \times 10^9) = 80$ ps, which corresponds to a Nyquist frequency of 6.25 GHz. The driver must therefore provide sufficient bandwidth well beyond this frequency to preserve edge rates and eye opening at the target data rate.

2. Hand Calculations — Tail Current and Load Resistor

The driver is a standard differential CML pair with resistive loads R_D , sized for 50 Ω single-ended (100 Ω differential) termination to match the floating receiver load. The tail current I_{SS} is set to deliver the required 600 mVppd differential swing directly into the 100 Ω load:

$$R_D = 50 \, \Omega \text{ (For matching)}$$

$$V_{swing, ppd} = 2 \cdot I_{SS} \cdot R_D / 2 = I_{SS} R_D$$

$$600 \, mV = I_{SS} \times 50 \, \Omega \rightarrow I_{SS} = \mathbf{12 \, mA}$$

The 12.5 Gb/s data rate maps to a Nyquist (fundamental) frequency of 6.25 GHz, which sets the minimum small-signal bandwidth the driver's output pole must clear with margin so that the eye is not excessively closed by ISI at the top bit rate.

3. Circuit Schematic

Figure 1 shows the CML driver circuit test bench used for both the AC and transient simulations: a differential pair with tail current source I_{SS} , resistive loads R_D (50 Ω per side), and the floating 100 Ω receiver termination connected between the differential outputs. An optional 400 fF single-ended output capacitor (C_{out}).

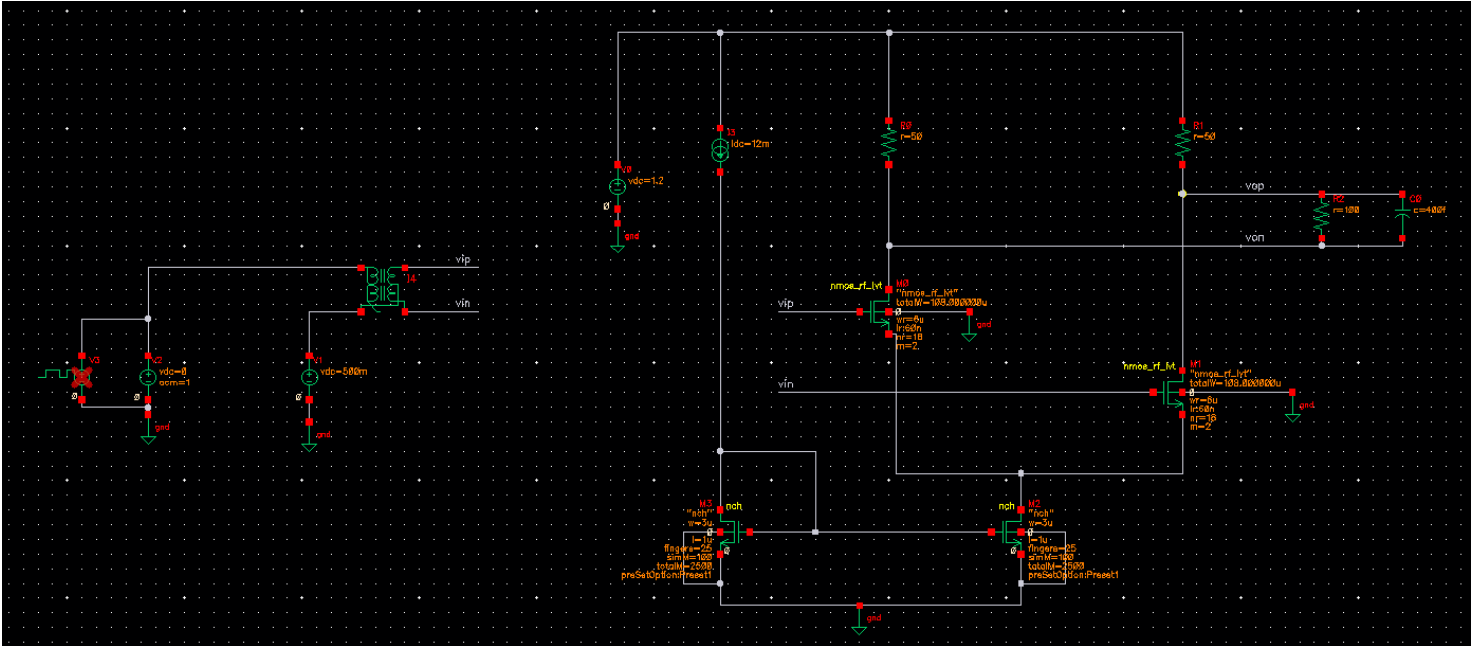


Figure 1 — CML driver test bench schematic (differential pair, RD loads, 100 Ω floating Rx load, switchable 400 fF output capacitors).

4. AC Analysis (1 GHz $\rightarrow$ 100 GHz)

With the PRBS source disabled and the AC source enabled, a small-signal AC sweep from 1 GHz to 100 GHz is run to extract the differential voltage gain A_v and the -3 dB bandwidth of the driver. The sweep is repeated twice — with and without the 400 fF output capacitor — to quantify the bandwidth penalty introduced by output loading.

4.1 Differential Gain A_v vs. Frequency

Axes: x-axis — Frequency (log scale, 1 GHz to 100 GHz); y-axis — Differential gain A_v .

With output capacitor $C_{out} = 400$ fF

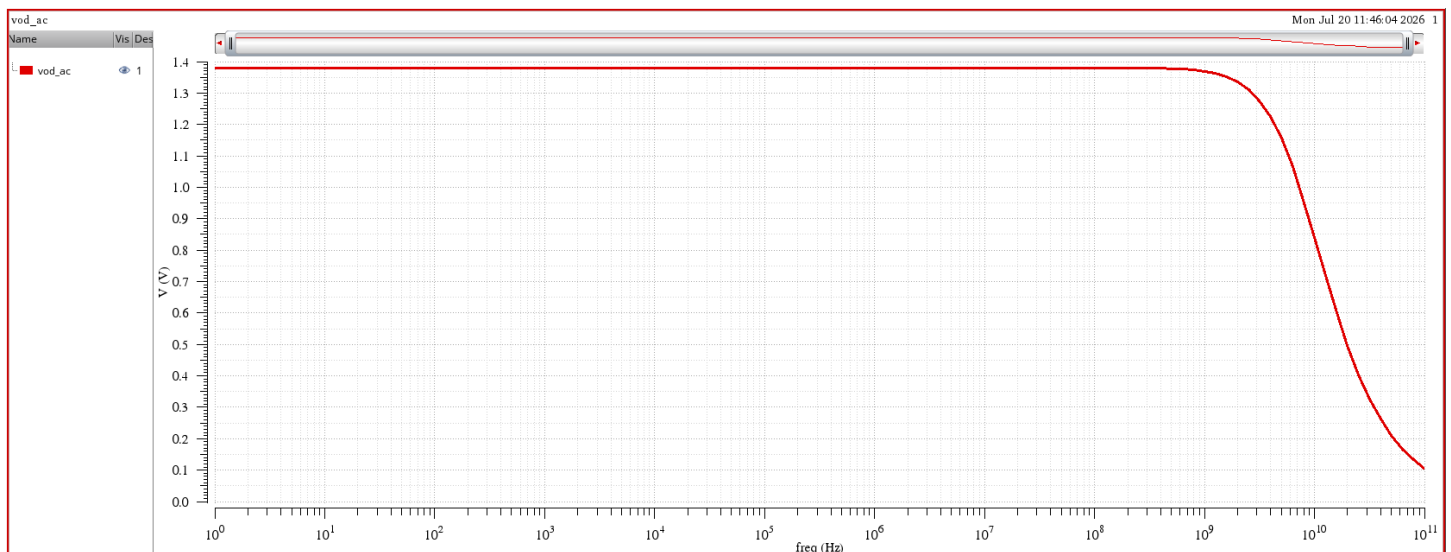


Figure 2 — Differential gain A_v vs. frequency, 1 GHz–100 GHz, with 400 fF output capacitors.

Without output capacitor

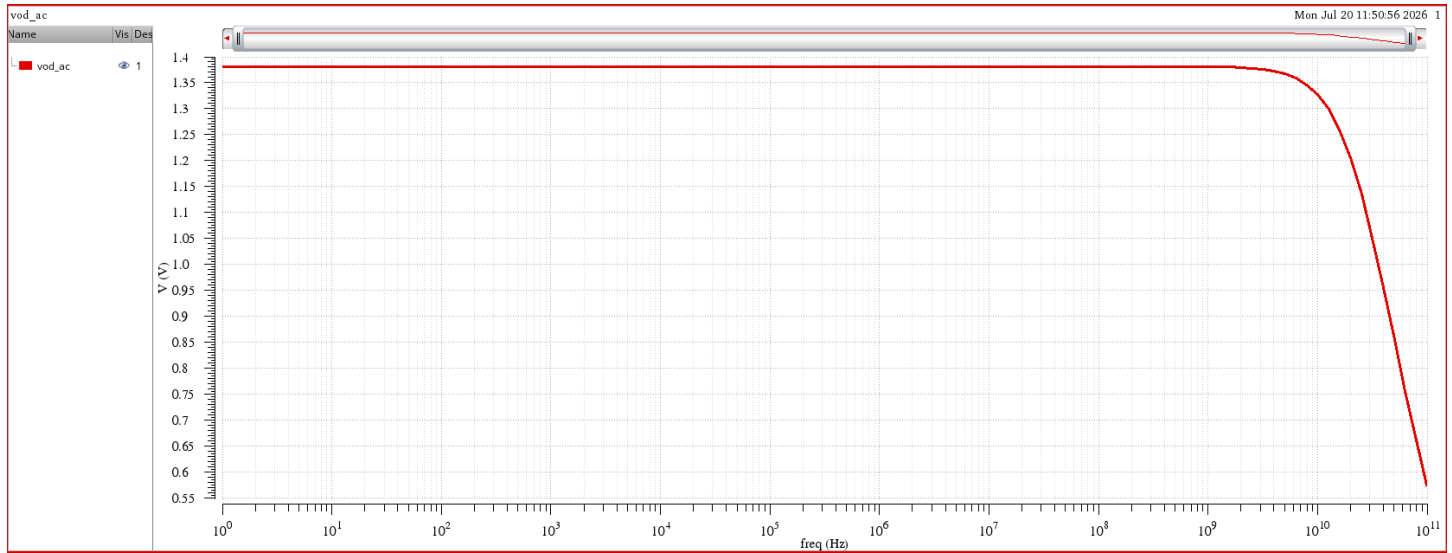


Figure 3 — Differential gain A_v vs. frequency, 1 GHz–100 GHz, without output capacitors.

With C_{out} removed, the only capacitance at the output node is the intrinsic drain/parasitic capacitance of the driver transistors and routing, which is much smaller than 400 fF. The gain therefore stays flat over a much wider frequency range before rolling off, i.e. the –3 dB bandwidth is pushed out to a substantially higher frequency than the loaded case.

4.2 3 dB Bandwidth Calculation

The dominant output pole is set by the parallel combination of the load resistance and the total capacitance at the output node (device drain capacitance C_{dd} plus, when enabled, the external C_{out} on each side):

$$f_{3dB} = \frac{1}{2\pi \left(\frac{RD}{2}\right) (C_{dd} + 2C_{out})}$$

The input-pair parasitic capacitance extracted from the operating-point/parametric analysis (used to evaluate C_{dd} above) is shown in Figure 4.

```
vds=826.2m
vth=366.6m
vdsat=87.6m
gmoverid=13.06
cdd=57.98f
```

Figure 4 — Input differential-pair device parameters used to extract C_{dd} for the bandwidth hand-calculation.

With output capacitor Cout = 400 fF

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
Assignments:CML_Driver:1	von_ac				
Assignments:CML_Driver:1	vod_ac				
Assignments:CML_Driver:1	vop_ac				
Assignments:CML_Driver:1	BW	7.677G			

Figure 5 — Bandwidth calculation with 400 fF output capacitors.

$$BW = \frac{1}{2\pi \cdot \left(\frac{50}{2}\right) \cdot (257.98 \text{ fF} + 800 \text{ fF})} \approx 7.42 \text{ GHz.}$$

This is comfortably above the 6.25 GHz Nyquist frequency required for 12.5 Gb/s operation, but with limited margin — consistent with the visibly earlier roll-off seen in Figure 2.

Without output capacitor

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
Assignments:CML_Driver:1	von_ac				
Assignments:CML_Driver:1	vod_ac				
Assignments:CML_Driver:1	vop_ac				
Assignments:CML_Driver:1	BW	37.92G			

Figure 6 — Bandwidth calculation without output capacitors.

5. Transient Analysis (80 ns, 1 ps max step)

With the AC source disabled and the PRBS source enabled, an 80 ns transient simulation (1 ps maximum time step) is run to verify large-signal behavior: correct 600 mVppd swing, symmetric rise/fall times, and clean single-ended/differential waveforms at both the input and output. As with the AC sweep, the simulation is repeated with and without the 400 fF output capacitor.

5.1 With Output Capacitor ($C_{out} = 400$ fF)

Input — single-ended

Axes: x-axis — time (0–80 ns, full transient window); y-axis — single-ended input voltage v_{ip} (V). Condition: PRBS input pattern driving the predriver, $C_{out} = 400$ fF.

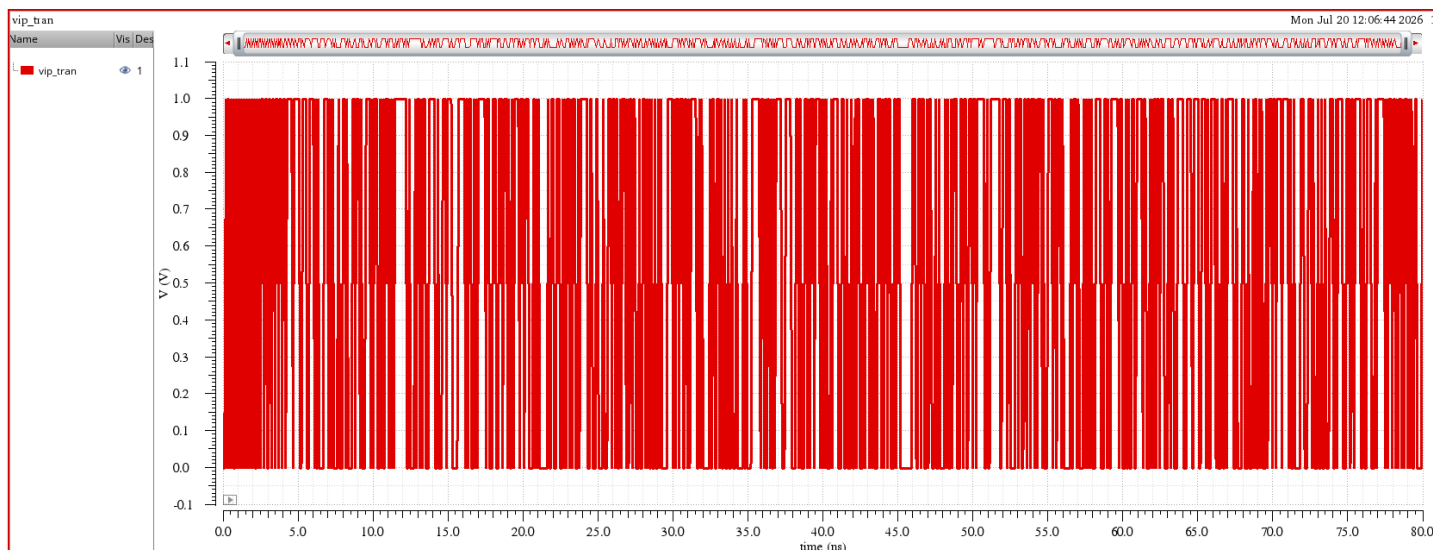


Figure 7 — Single-ended input voltage, full 80 ns transient window.

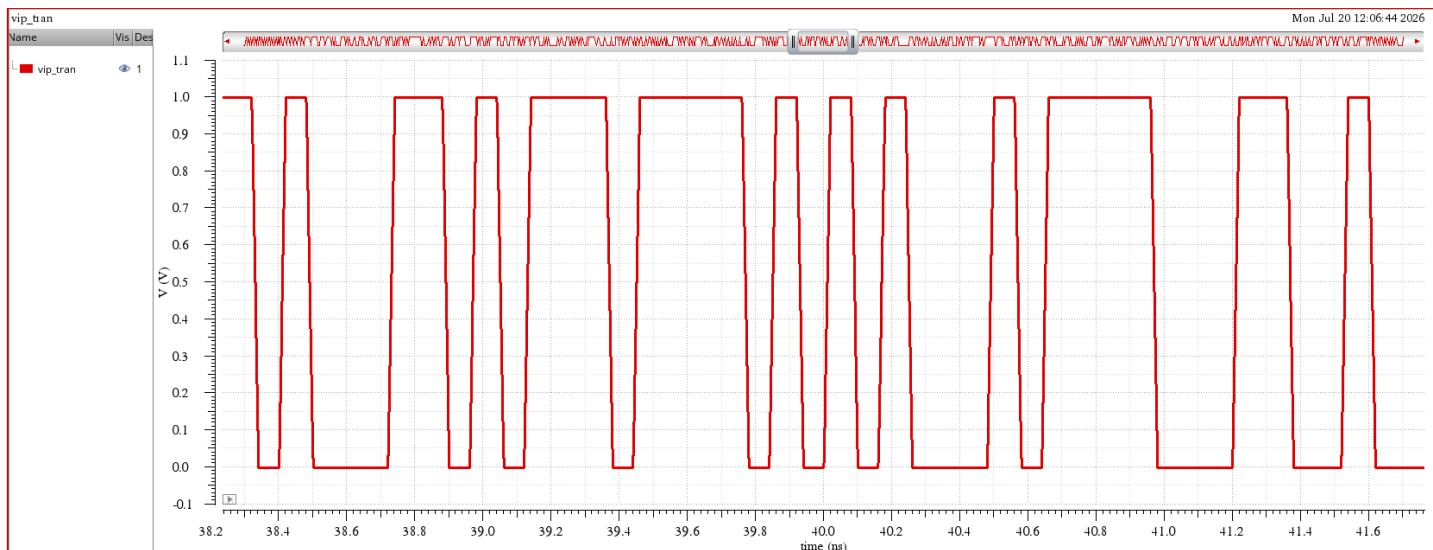


Figure 8 — Single-ended input voltage, zoomed strip (a few unit intervals) showing PRBS transitions and 80 ps UI edges.

The single-ended input toggles between its high/low logic levels following the PRBS pattern, with sharp, well-defined 80 ps unit intervals. The zoomed strip in Figure 8 confirms the input edge rate is fast relative to the UI, as required to drive the CML pair cleanly into switching.

Input — differential

Axes: x-axis — time (ns); y-axis — differential input voltage $v_{ip}-v_{in}$ (V). Condition: $C_{out} = 400$ fF.

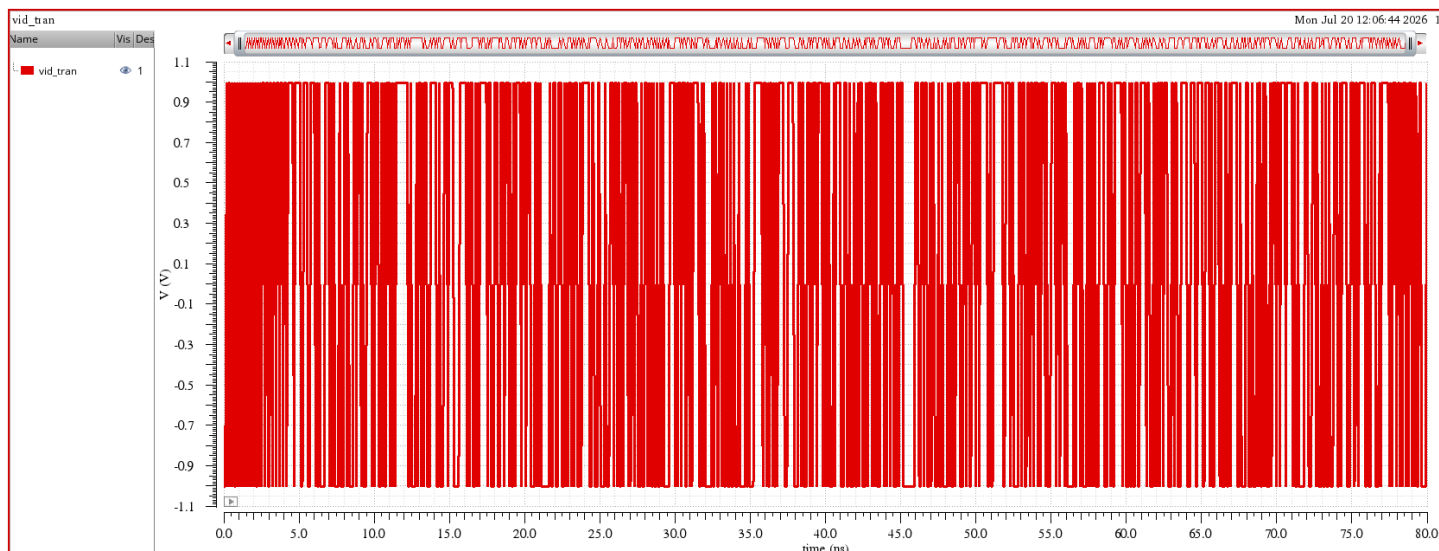


Figure 9 — Differential input voltage, full transient window.

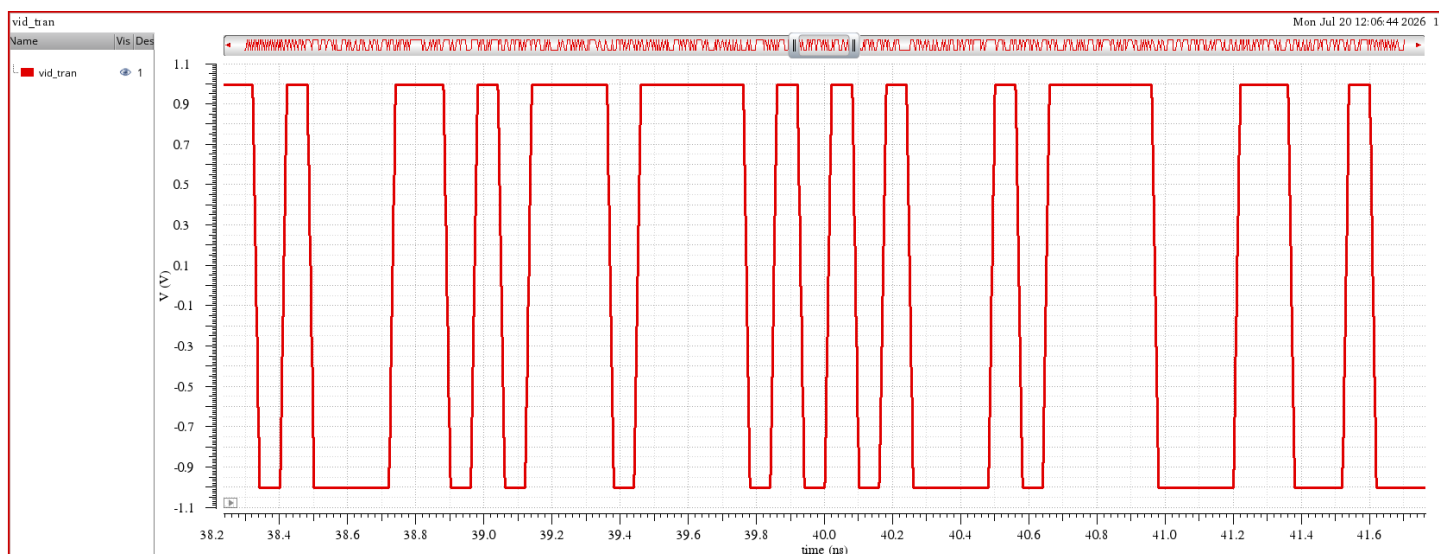


Figure 10 — Differential input voltage, zoomed strip.

The differential input swings symmetrically about 0 V between its two logic levels, confirming balanced differential drive into the CML pair with no significant common-mode input offset.

Output — single-ended

Axes: x-axis — time (ns); y-axis — single-ended output voltage v_{op} (V) at the $100\ \Omega$ floating load. Condition: $C_{out} = 400\text{ fF}$.

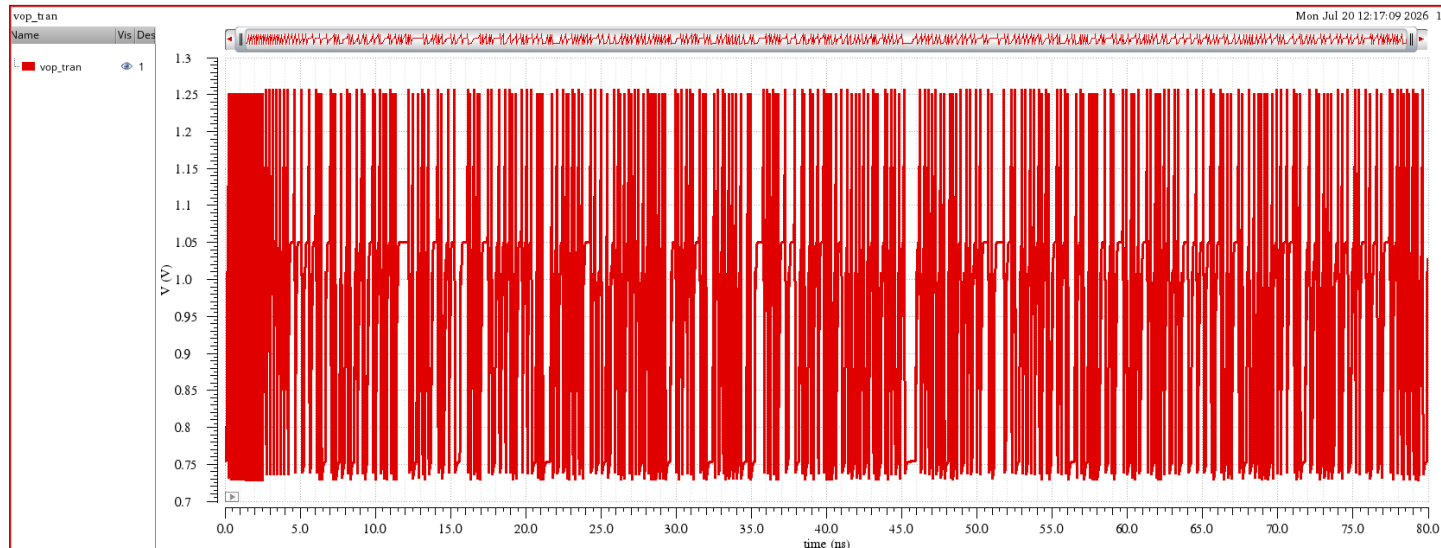


Figure 11 — Single-ended output voltage, full transient window, with 400 fF output capacitors.

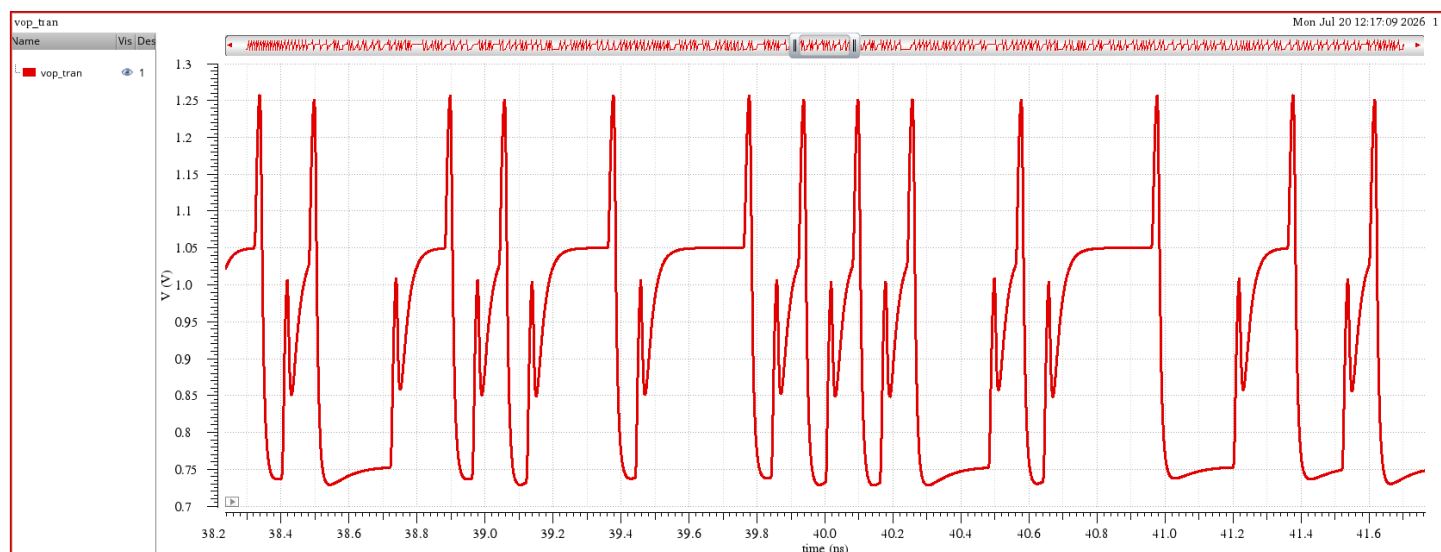


Figure 12 — Single-ended output voltage, zoomed strip, with 400 fF output capacitors.

Each output swings between V_{DD} and $V_{DD} - I_{ssRD}$, giving a single-ended swing of 300 mVpp per side (600 mVppd differential).

Output — differential

Axes: x-axis — time (ns); y-axis — differential output voltage $v_{op}-v_{on}$ (V). Condition: $C_{out} = 400$ fF.

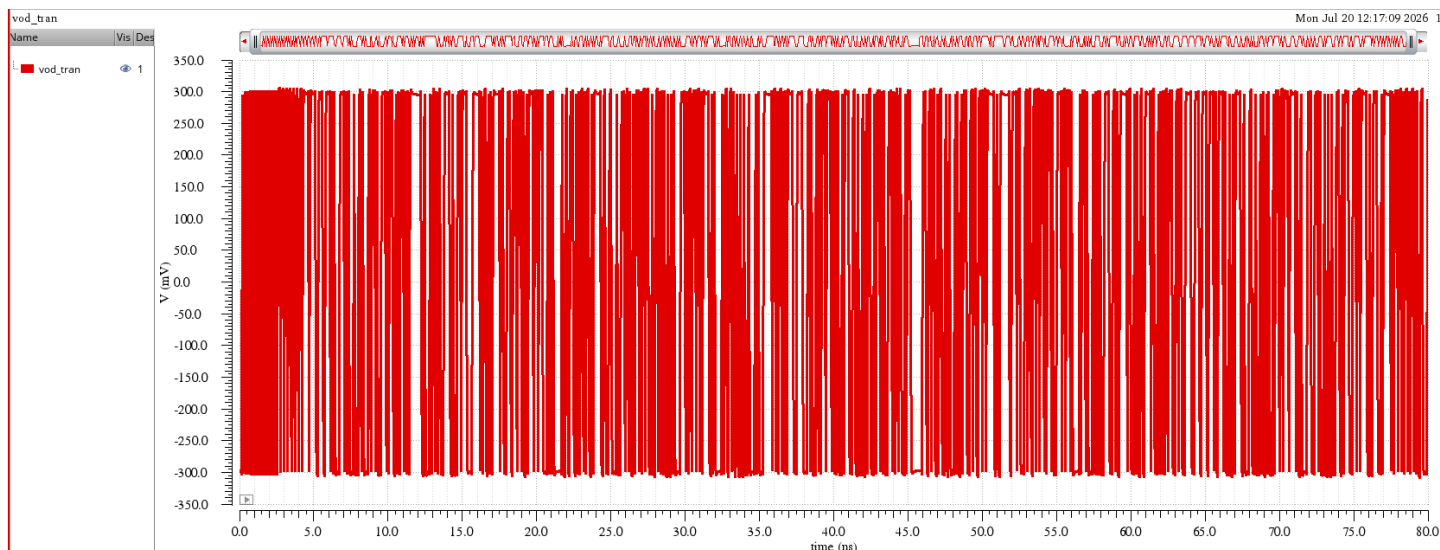


Figure 13 — Differential output voltage, full transient window, with 400 fF output capacitors.

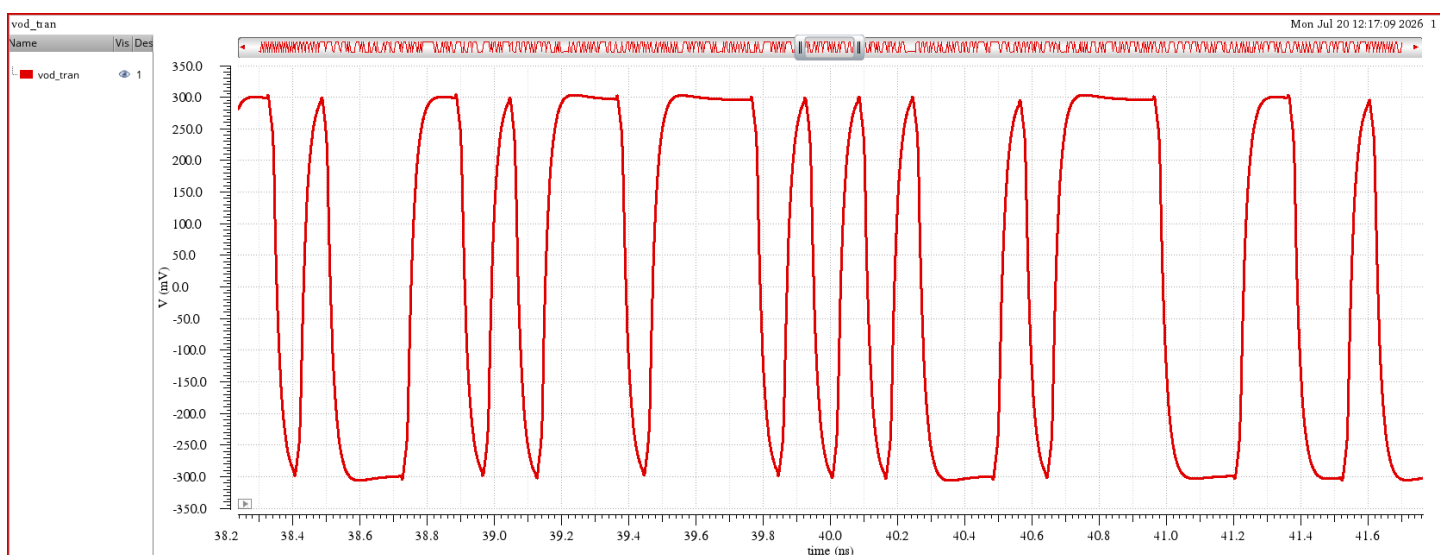


Figure 14 — Differential output voltage, zoomed strip, with 400 fF output capacitors.

The differential output reaches the target ± 300 mV (600 mVppd) swing at the $100\ \Omega$ load, confirming the I_{SS}/R_D sizing from Section 2 is correct.

5.2 Without Output Capacitor

The same set of waveforms is repeated with C_{out} removed. The input waveforms (Figures 7–10, driven by the same PRBS source and predriver) are unchanged, since the output capacitor does not load the input nodes.

Output — single-ended

Axes: x-axis — time (ns); y-axis — single-ended output voltage vop/von (V). Condition: no output capacitor.

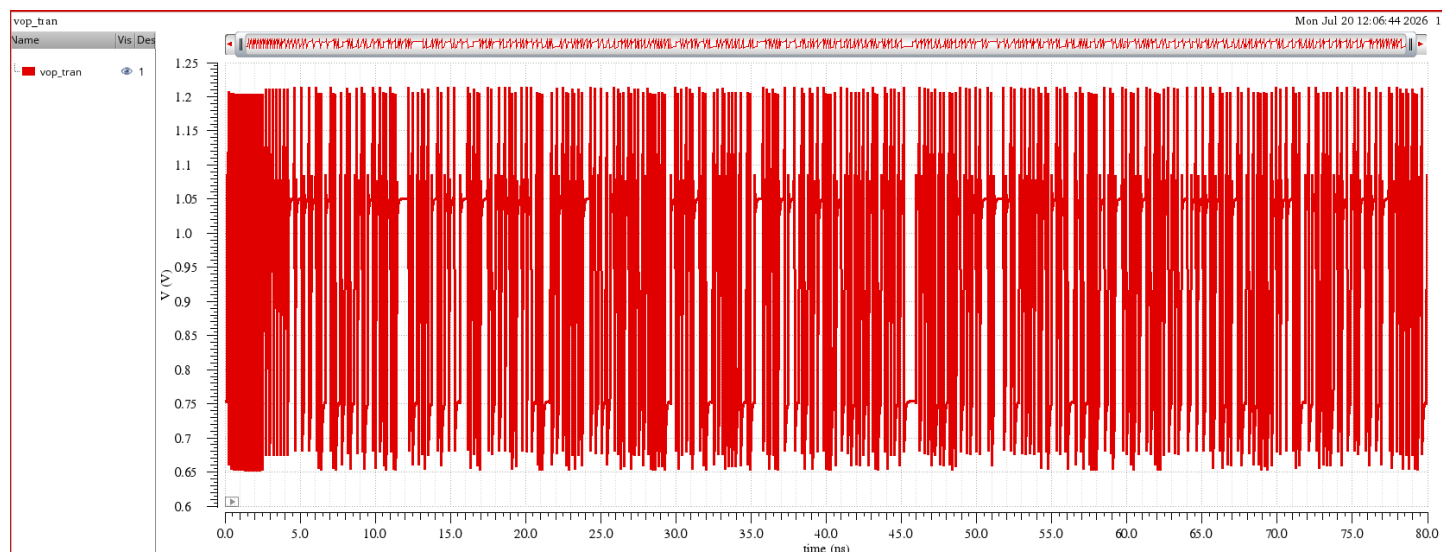


Figure 15 — Single-ended output voltage, full transient window, without output capacitors.

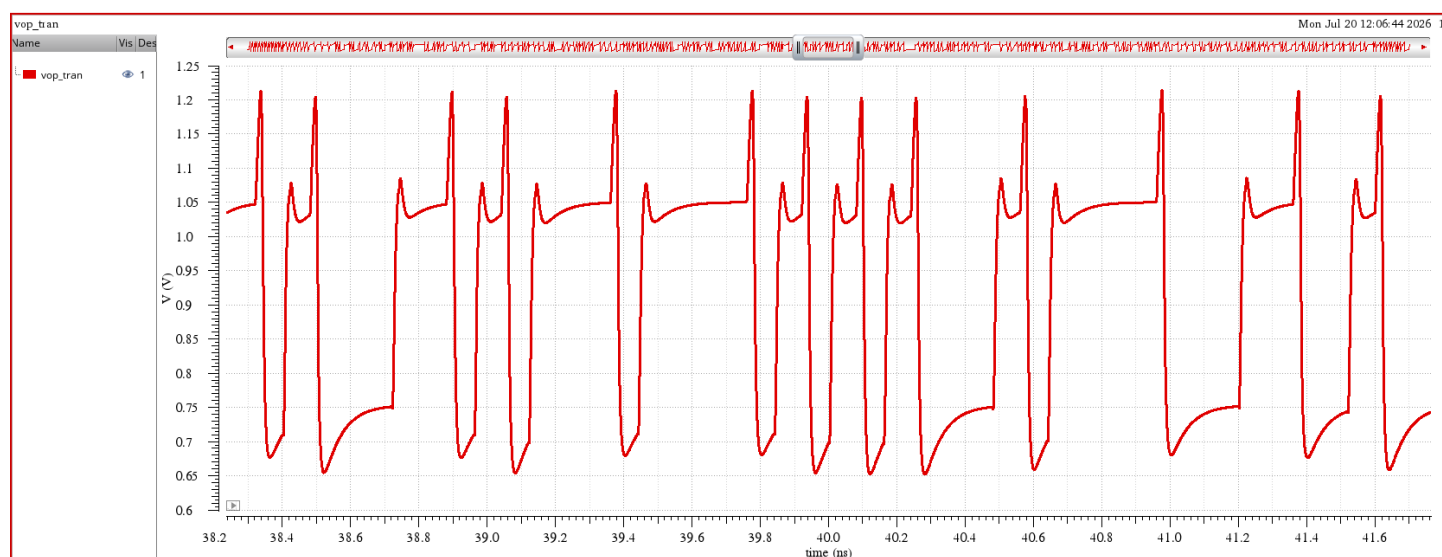


Figure 16 — Single-ended output voltage, zoomed strip, without output capacitors.

With C_{out} removed, the output edges in the zoomed strip (Figure 16) are visibly sharper than the loaded case (Figure 12), directly reflecting the much higher bandwidth calculated in Section 4.2. The 300mV single-ended swing amplitude is unchanged, as expected since I_{ss} and R_D were not altered.

Output — differential

Axes: x-axis — time (ns); y-axis — differential output voltage $v_{op}-v_{on}$ (V). Condition: no output capacitor.

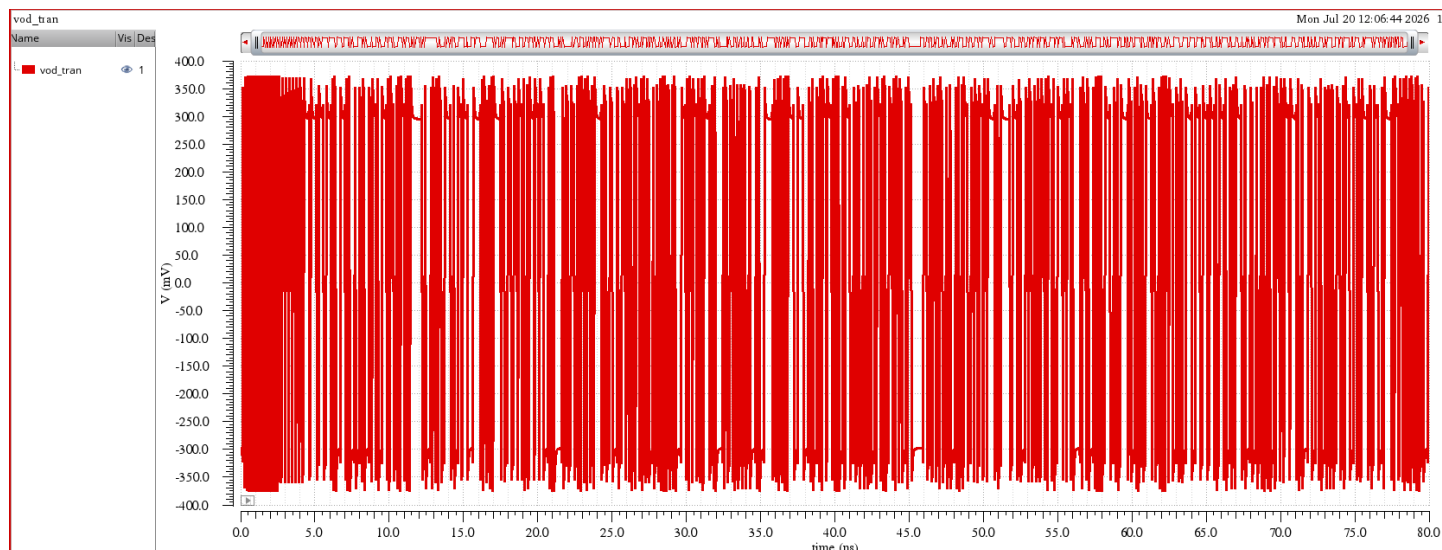


Figure 17 — Differential output voltage, full transient window, without output capacitors.

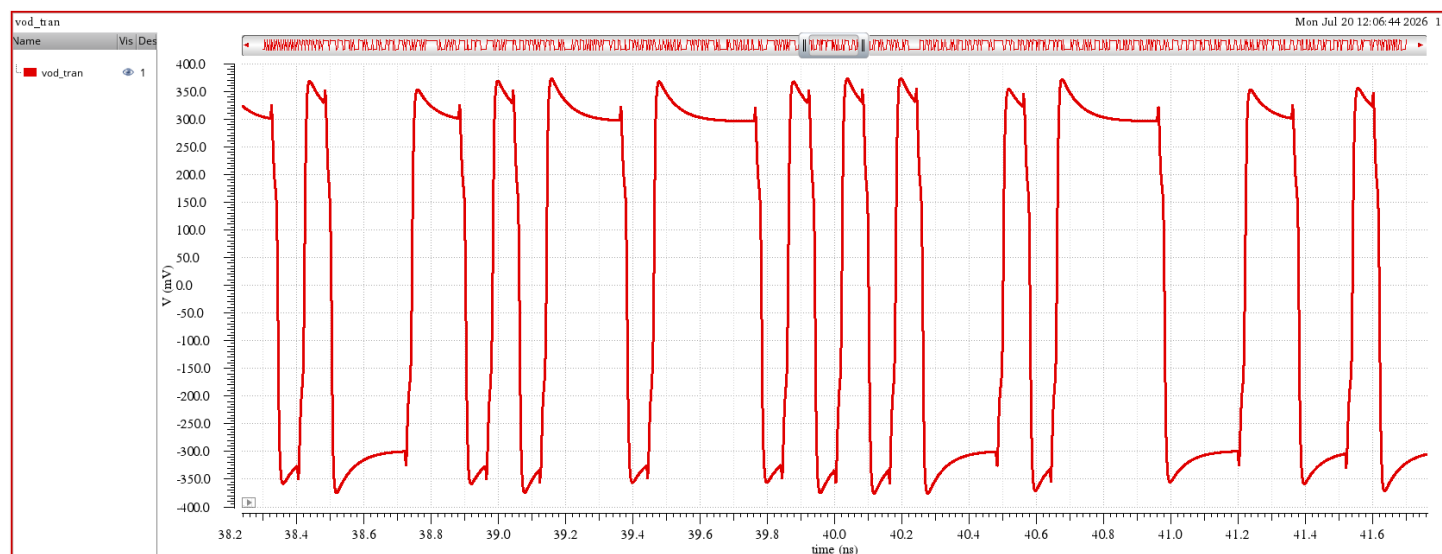


Figure 18 — Differential output voltage, zoomed strip, without output capacitors.

The differential output again settles to the full ± 300 mV swing, but with faster, cleaner 0-to-100 % transitions than the capacitor-loaded case.

6. Eye Diagram

The eye diagram is generated from the differential output waveform $vop - von$ using:

```
eyeDiagram( vtime('tran "/vop") - vtime('tran "/von"), 100*80p, 80n, 2*80p, ?autoCenter t )
```

i.e. the differential output is folded over a 2-UI (160 ps) window at a 100×80 ps sampling resolution across the full 80 ns transient run, auto-centered on the optimal sampling instant. This is the standard visualization for checking eye height, eye width, jitter, and overall link margin against the 80 ps UI target.

With output capacitor ($C_{out} = 400$ fF)

Axes: x-axis — time within a 2-UI window (≈ 160 ps); y-axis — differential output voltage $vop-von$ (V).

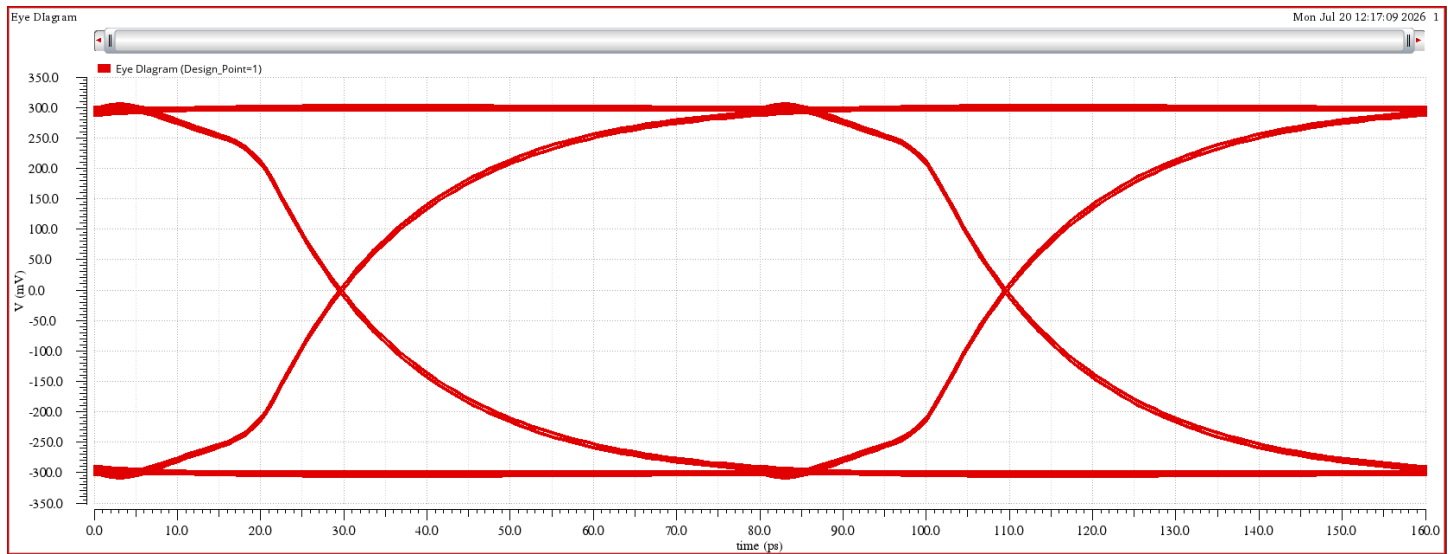


Figure 19 — Eye diagram of the differential output, with 400 fF output capacitors.

The eye is open at the center of the UI, with the vertical opening approaching the full ± 300 mV differential swing. Because of the reduced bandwidth from the output loading, the crossing points are visibly less sharp and the eye is somewhat narrower/rounder at the edges — a direct consequence of the slower rise/fall times observed in Section 5.1. Even so, the eye remains open with reasonable margin at 12.5 Gb/s.

Without output capacitor

Axes: x-axis — time within a 2-UI window (≈ 160 ps); y-axis — differential output voltage $v_{op}-v_{on}$ (V).

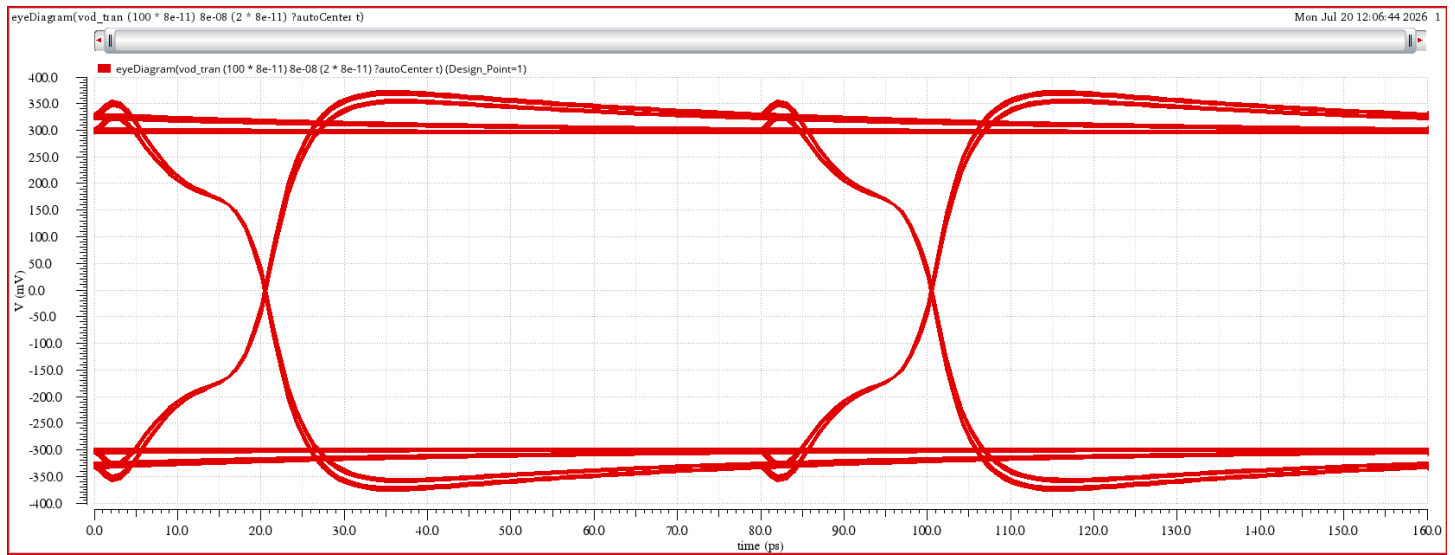


Figure 20 — Eye diagram of the differential output, without output capacitors.

With C_{out} removed, the eye is visibly wider and more rectangular, with sharper crossing transitions and a larger effective eye opening in both height and width compared to Figure 19. This confirms that the 400 fF output capacitance is the primary source of eye closure/ISI in this design, and that eliminating (or minimizing) parasitic/package capacitance at the driver output is the single most effective lever for improving eye margin at 12.5 Gb/s.

7. Conclusion

A CML NRZ driver was designed with $I_{ss} = 12$ mA and $R_D = 50$ Ω per side to deliver the required 600 mVppd differential swing into a 100 Ω floating receiver load. Small-signal AC analysis confirmed a dominant single-pole response at the output, with -3 dB bandwidth of ≈ 7.42 GHz when a realistic 400 fF output capacitance is included — sufficient to clear the 6.25 GHz Nyquist frequency of a 12.5 Gb/s NRZ link, though with tighter margin than the unloaded case. Transient simulation confirmed correct DC swing and PRBS operation at both the input and output, and the resulting eye diagrams were open under both loading conditions, with the output capacitor visibly reducing eye width and edge sharpness as expected from the bandwidth calculation. Overall, the driver meets the target DC-coupled link specification, with output/package capacitance identified as the dominant factor limiting bandwidth and eye margin — and therefore the primary target for further optimization (e.g., inductive peaking or reduced parasitic loading) if additional margin is required.

Appendix — Expressions Used

Name	Type	Details
von_ac	expr	mag(v("/von" ?result "ac"))
vod_ac	expr	mag((v("/vop" ?result "ac") - v("/von" ?result "ac")))
vop_ac	expr	mag(v("/vop" ?result "ac"))
BW	expr	bandwidth(vod_ac 3 "low")
vop_tran	expr	v("/vop" ?result "tran")
von_tran	expr	v("/von" ?result "tran")
vod_tran	expr	(vop_tran - von_tran)
vip_tran	expr	v("/vip" ?result "tran")
vin_tran	expr	v("/vin" ?result "tran")
vid_tran	expr	(vip_tran - vin_tran)
Eye Dliagram	expr	eyeDiagram(vod_tran (100 * 8e-11) 8e-08 (2 * 8e-11) ?autoCenter t)

Figure 21 — Summary of expressions used for gain, bandwidth, and eye-diagram generation.